BetaNav / BetaModel

LightNav token prediction + a separate conditional flow-matching action expert

CE-trainable · FM-trainable · Frozen / fixed

Current training objective: L = L_CE + 2.0 x L_FM

A SHARED PERCEPTION + TOKEN SUPERVISION

Observed RGB history
Frames at or before t
470 x 280 pixels (W x H)

SlowFast + resize
Current / Fast / Mid / Long / Anchor
448 x 256 pixels (W x H)

Instruction + prompt
Tokenizer / text embeddings

Qwen3-VL vision tower
Frozen ViT + visual merger

Post-ViT token pooling
Tier-dependent spatial pooling

text tokens

LightNav / Qwen3-VL
Pretrained causal text backbone
Hidden width D = 2,560
One joint forward pass
Training: prompt + GT answer

LM head

Next-token predictions
<apos_k> <opos_j>
<act_l0_a> <act_l1_b> <act_l2_c>
+ end-of-answer token

CE loss
Masked next-token CE

Token GT + mask
RVQ: 3 codes / 10-step chunk
Invalid labels = -100

CE updates the text backbone + LM head

Prefix hidden states
Before answer_start; non-padding only
[B, N, 2,560] | no GT answer features

STOP GRADIENT
hidden.detach()

B FLOW-MATCHING TRAINING

Continuous action GT
a_raw: local SE(2) increments [B,30,3]
30 future recorded steps, before RVQ
Validity mask [B,30]

Normalize: a = a_raw / scale

Sample noise + flow time
epsilon ~ N(0,I) tau ~ U(0,1)
x_tau = (1 - tau) epsilon + tau a
Target flow: v* = a - epsilon
Invalid future steps: noise, no loss

context [B,N,512]

FlowActionExpert | ~36.5M parameters
Action Linear 3 -> 512 + learned step embeddings

Self-attention	Cross-attention	FFN
Bidirectional actions	K,V = projected prefix	512 -> 2,048 -> 512

6 blocks | 8 heads | tau: sinusoidal embedding + MLP -> FiLM

Context projection
Linear 2,560 -> 512

Predicted flow v_theta
Linear 512 -> 3 | [B,30,3]

FM loss
Masked MSE(v_theta, a - epsilon)

FM gradients update only the context projection + action expert

C INFERENCE | model.predict(...)

Prompt-only prefill
Cache prefix hidden states
Reuse the same context for FM

Generate tokens + RVQ decode
3 RVQ tokens -> 10-step RVQ path
Explicit STOP or near-zero RVQ?

STOP: zero path; skip FM

CONTINUE: run FM expert

Noise -> normalized action chunk
10 Euler integration steps, tau: 0 -> 1
x_next = x + v_theta(x, tau, context) / 10
Output: 30 future actions (not 10)

Continuous trajectory
Denormalize -> SE(2) composition
30 local increments + 30 waypoints
[forward m, left m, yaw rad]

Implementation: src/lightnav/inference/beta_model.py | FM predicts flow-time derivatives; executed outputs are SE(2) increments.